Aarush Kandukoori

aarushkandukoori.com | akanduko@andrew.cmu.edu | github.com/aarushkandukoori | linkedin.com/in/aarush-kandukoori

Education

Carnegie Mellon University Pittsburgh, PA

B.S. Computer Science, Statistics & Machine Learning *Aug. 2025 – May 2028*

Coursework: Introduction to Probability Theory, Matrices & Linear Transformations, Methods for Statistics & Data Science, Concepts of Mathematics, Calculus in Three Dimensions, Fundamentals of Programming & Computer Science, Art & Machine Learning, Independent Study in Computer Science

Experience

BUMP Pittsburgh, PA

Co-Founder & Chief Executive Officer *Sept. 2023 – Present*

- Lead a 3-person team building a patent-pending closed-loop wearable that detects sustained bradycardia and autonomously delivers atropine (USPTO provisional #63/686,176).
- Wrote the real-time detection software: a streaming Pan-Tompkins QRS detector with **97.7%** sensitivity and **99.4%** positive predictive value across **44** non-paced MIT-BIH records, feeding a deterministic sustained-bradycardia alert rule; **4 Docker** application services plus Redis Streams and TimescaleDB, inside a **250 ms** processing budget enforced by an integration test.

Yale University, Computer Science (Prof. Lin Zhong's group) Remote

Research Intern, On-Device LLM Serving (Pie) *Aug. 2026 – Present*

- Ported **Pie**, Yale's programmable LLM serving system (SOSP '25), to iOS: routed WebAssembly inferlets through **Wasmtime's Pulley interpreter** to satisfy Apple's no-JIT rule and added a file-backed mmap shared-memory fallback for the iOS sandbox.
- Brought it up on a physical **iPhone 16 Pro** by fixing four failures the Simulator never showed, including a Wasmtime pooling allocator reserving 4 TB of virtual address space; **Qwen3-0.6B** now runs on the phone at ∼**60 tok/s** with **0.57 s** time to first token and **1.0 GiB** peak memory.

Carnegie Mellon University, Database Group Pittsburgh, PA

Undergraduate Researcher, Query Engines & Storage *May 2026 – Present*

- Extended PhD mentor Drew Ripberger's vertical-partitioning study (one **Apache Parquet** table split column-wise across 2–3 files and queried as one in **Apache DataFusion**) from TPC-H to all **99** TPC-DS queries.
- Added an S3/MinIO object-store backend with per-request counting: partitioned layouts ran **8–13%** slower by geometric mean (TPC-DS 1.08x; 25 queries faster, 57 slower), and S3 GET requests rose **22–67%** with bytes read flat.
- Traced a **7.2x** Q39 slowdown to zipped scans receiving no dynamic-filter pushdown (11.7M rows read vs. 245K), and found 18 hot columns hold only **6.6%** of bytes, supporting the hot/cold split Drew proposed; summarized results in a 3-page report (Sept. 2026).

Harvard Medical School & Brigham and Women's Hospital Boston, MA

Research Intern, Ann Romney Center for Neurologic Diseases *July 2024 – Aug. 2024*

- Youngest intern in the lab's history; performed gated multicolor flow cytometry, qPCR analysis, and RNA/DNA extraction with PCR to study T cells for the lab's multiple sclerosis research, and supported in vivo mouse trials.

Stanford University, Kevin Alexander Lab Stanford, CA

Data Analytics & Research Intern, Cardiovascular Institute *June 2024 – Aug. 2024*

- Engineered **FibroQuant**, a Java and ImageJ macro tool for biomedical image analysis, processing ∼**300** images from **15** mice at **2x** prior throughput.

University of Maryland, MIND Lab College Park, MD

Software Engineering & Data Science Intern *Summers 2022 & 2023*

- Wrote Python scripts to plot, label, and filter Spire Tag breathing-sensor data, categorizing **450+** breathing segments as inverted or noisy.

Projects

CoT Faithfulness Replication Protocol | *Python, Transformers, LLM-as-Judge* github.com/aarushkandukoori/AI-Research-CoT-Project

- Built a replication protocol and pipeline for the Turpin et al. (2023) hint-injection measure of chain-of-thought faithfulness on BIG-Bench Hard with Qwen2.5: inject biasing hints, isolate the answer flips they cause, and check whether the stated reasoning admits the hint.
- Validated the pipeline on a synthetic cache with known ground truth; it failed to recover it, exposing **6** measurement-validity defects (including a judge that labeled every case faithful); co-authored with Aditya Kandukoori and reviewed in the ACL Rolling Review August 2026 cycle.

NBA Referee-Crew Market Backtest | *Python, LightGBM, Walk-Forward CV* github.com/aarushkandukoori/nba-referee-market-edge

- Tested whether referee-crew tendencies create a tradable edge on game totals across **5,041** games, **1,838** of them scored out-of-sample, using rolling crew features, shrinkage toward a time-varying league prior, and the market line as a control.
- Reported a null result: crew features worsened out-of-sample log-loss by 0.00038 (permutation p = 0.57), with look-ahead bias designed out via season walk-forward splits, pre-tip-only pricing, and a dedicated leakage test suite.

Publications & Patents

- Aarush Kandukoori, Aditya Kandukoori, Faizan Wajid. *Comparative Analysis of Digital Tools and Traditional Teaching Methods in Educational Effectiveness.* arXiv preprint, arxiv.org/abs/2408.06689 (2024). 1st author.
- U.S. Provisional Patent #63/686,176: wearable closed-loop atropine delivery system for crashing bradycardia (BUMP).

Technical Skills

- **Languages:** Python, Rust, Swift, C++, Java, SQL, TypeScript, JavaScript, R, Bash
- **ML & Data:** PyTorch, MLX, Core ML, ggml, scikit-learn, LightGBM, NumPy, pandas, Hugging Face Transformers, LLM evaluation & benchmarking
- **Systems & Web:** Apache DataFusion, DuckDB, Apache Parquet, Docker, Redis Streams, TimescaleDB, WebAssembly (Wasmtime), iOS/SwiftUI, Git